Prediction Pipeline of Rare Genetic Diseases

MSc in Bioinformatics for Computational Genomics
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1 ABSTRACT

This project aims to identify causal genetic variants for rare Mendelian diseases in five family trios using simulated Whole Exome Sequencing (WES) data focused on human chromosome 20 (GRCh38). A standardized bioinformatics pipeline was implemented in a Unix environment, encompassing raw read quality control (**FastQC**), sequence alignment (**Bowtie2**), variant calling (**Freebayes**), and clinical annotation (**Ensembl VEP**). Variants were prioritized based on their functional impact (**IMPACT: HIGH/MODERATE**), population frequency (**MAX_AF < 0.0001**), and inheritance models (**Autosomal Dominant and Recessive**). The analysis successfully identified diagnostic variants in three cases: a frameshift mutation in *JAG1* (Trio 2), a frameshift in *GSS* (Trio 4), and a nonsense mutation in *MKKS* (Trio 5). Trios 1 and 3 were identified as negative controls, consistent with the expected simulation parameters.

2 INTRODUCTION

Rare diseases affect approximately 3.5% – 5.9% of the global population, often manifesting in early childhood with severe phenotypes. About 80% of these conditions have a genetic basis, primarily following Mendelian inheritance patterns. Due to high genetic heterogeneity, **Whole Exome Sequencing (WES)** has become a primary diagnostic tool, focusing on the ~1.5% of the genome that contains roughly 85% of known disease-causing mutations.

This report documents the bioinformatics analysis of 5 trios (offspring and parents). The simulation utilizes real genetic backgrounds from the 1000 Genomes Project. The objective is to distinguish pathogenic mutations from benign polymorphisms, specifically focusing on **Chromosome 20**, a region rich in genes associated with multi-organ syndromes such as Alagille syndrome and McKusick-Kaufman syndrome.

3 MATERIALS AND METHODS

Data Availability and Reproducibility

To ensure the reproducibility of this diagnostic workflow, all scripts developed for the analysis (Bash scripts for the pipeline and awk/bcftools filtering commands) have been versioned and made publicly available.

- **Scripts and Documentation:** The complete source code, command logs, and final VCFs are available in the GitHub repository
- **Intermediate Files:** Raw data and intermediate files (bam, bai, fa, fa.fai, fq) are stored on the **leon** server at the address 159.149.160.7.

Computational Environment and Data

The analysis was performed on the **leon** server. The data consisted of paired-end FASTQ files. The reference genome used was **GRCh38** (Chromosome 20).

3 MATERIALS AND METHODS

Bioinformatics Pipeline

1. **Quality Control:** Raw reads were evaluated with **FastQC**. Reports were aggregated using **MultiQC**.
2. **Alignment:** Reads were mapped to the reference using **Bowtie2**. SAM files were converted to BAM, sorted, and indexed with **Samtools**.
3. **Post-Alignment QC: Qualimap BamQC** was used to calculate coverage statistics over the target regions defined by the chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed file.
4. **Variant Calling:** Joint genotyping was performed with **Freebayes** using a minimum coverage threshold of 20x.

Annotation and Filtering: Variants were annotated using **Ensembl VEP**. Prioritization was carried out via filter_vep and custom bcftools queries.

Implementation (Example Commands)

The complete pipeline is avaible on GitHub repository

→ Ciclo for per allineare i file fq dei 3 individui assegnando automaticamente i Read Groups:

```
for f in HG00421 HG00422 HG00423; do
    # Definiamo il ruolo in base al codice HG
    if [ "$f" == "HG00421" ]; then SM="child_trio_x"; fi
    if [ "$f" == "HG00422" ]; then SM="father_trio_x"; fi
    if [ "$f" == "HG00423" ]; then SM="mother_trio_x"; f
    bowtie2 -p 4 \
    -x /progetto/chr20 \
    -1 /progetto/cartella.dati/trio_x/${f}.targets_R1.fq.gz
\
    -2 /progetto/cartella.dati/trio_x/${f}.targets_R2.fq.gz
\
    --rg-id "$f" --rg "SM:$SM" | \
    samtools view -Sb - | \
    samtools sort -o ${SM}_sorted.bam
done
```

→ Filtraggio per modelli di ereditarietà:

L'ordine dei campioni nel VCF è fondamentale (es. 0=Mother, 1=Father, 2=Child)

```
grep "#CHROM" trio_x_annotated.vcf
```

CASO A: Autosomica Recessiva (AR)

```
bcftools view -i 'GT[2]="AA" && GT[0]="RA" &&
GT[1]="RA"' trio_x_annotated.vcf >
trio_x_genotipo.vcf
```

CASO B: Autosomica Dominante (AD) de novo

```
bcftools view -i 'GT[2]="RA" && GT[0]="RR" && GT[1]="RR"'
trio_x_annotated.vcf >
trio_x_genotipo.vcf
```

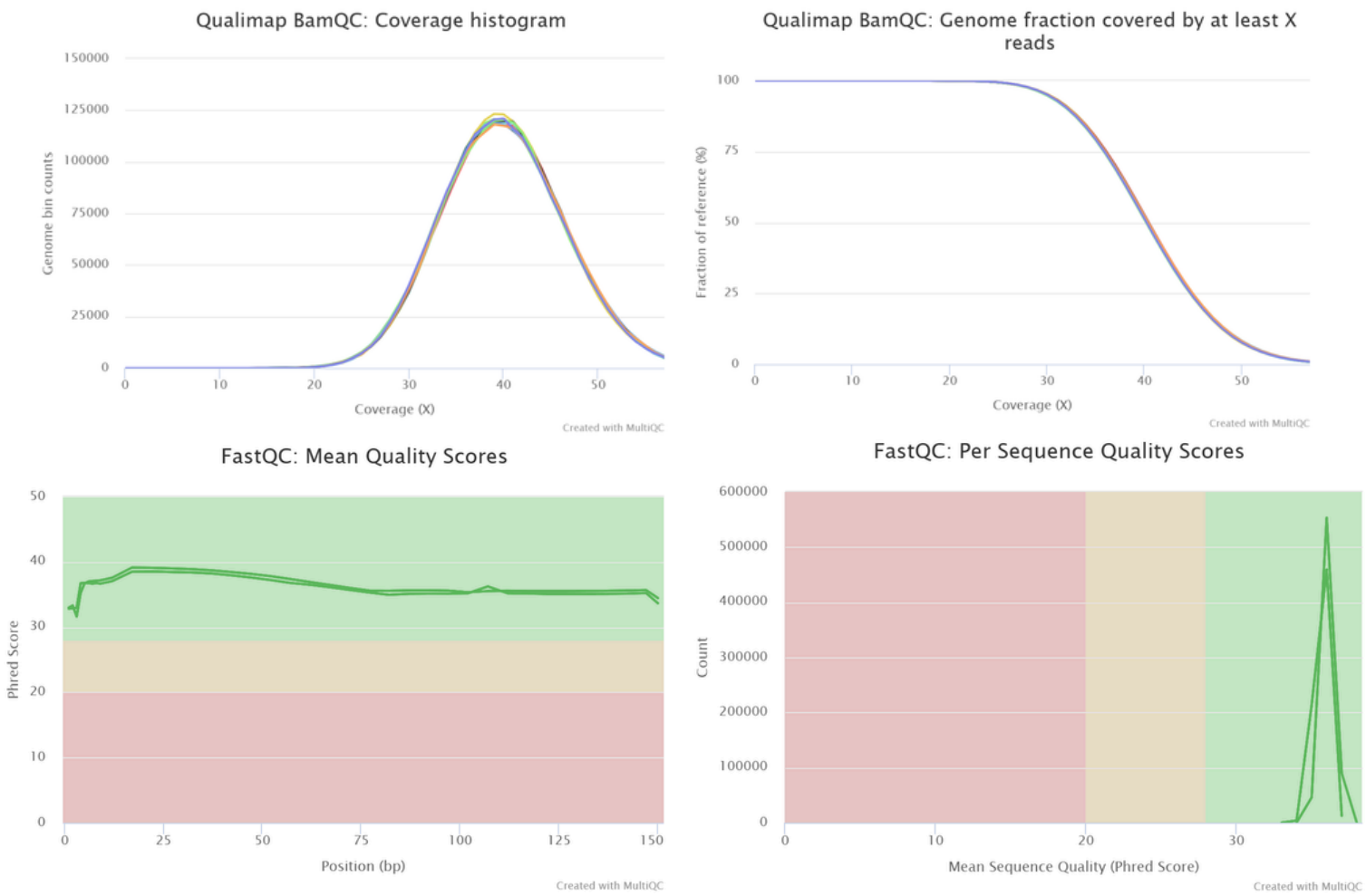
CASO C: Autosomica Dominante (AD) ereditata dal padre

```
bcftools view -i 'GT[2]="RA" && GT[1]="RA" && GT[0]="RR"'
trio_x_annotated.vcf >
trio_x_genotipo.vcf
```

4 RESULTS AND DISCUSSION

Quality Control and Coverage

FastQC analysis revealed high-quality sequencing data (Phred scores > 30 for most reads). Alignment rates exceeded 98% for all samples. MultiQC reports confirmed that while coverage was lower in some target regions, it remained sufficient (>20x) to proceed with variant calling according to project guidelines.



Summary of quality statistics for all trios

Diagnostic Results

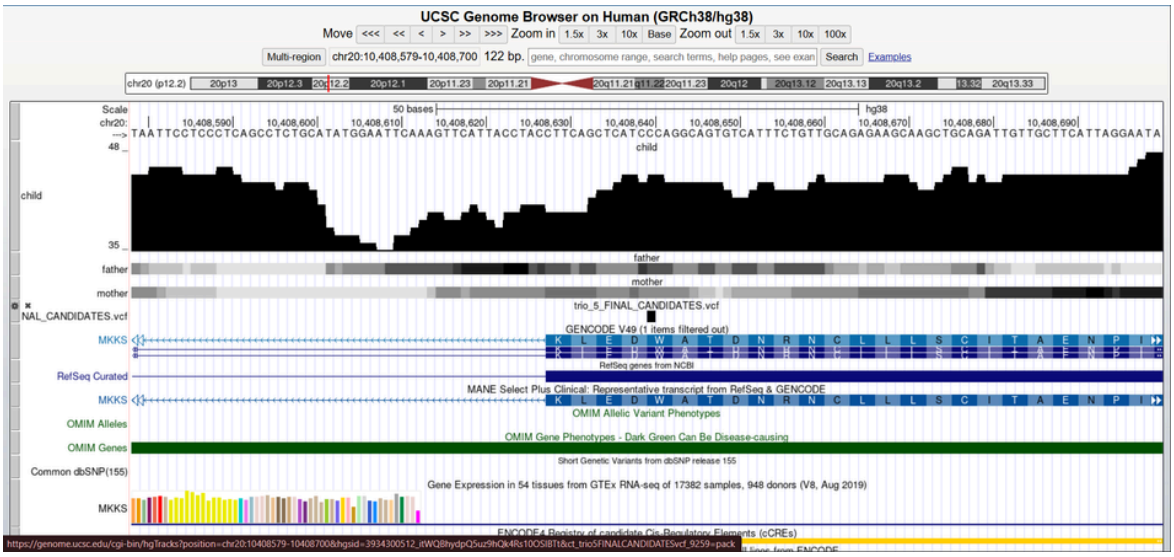
The following tables summarizes the molecular diagnosis for the assigned trios

Trio	Diagnosis	Gene	Inheritance
1	Healthy (Polymorphism)	SIRPA	-
2	Alagille Syndrome	JAG1	AD
3	Healthy (Negative)	-	-
4	GSS Deficiency	GSS	AR
5	McKusick-Kaufman Syndrome	MKKS	AR

Trio	Variant (HGVSc/p)	Consequence	Impact
1	p.Asn120Ser (various)	Missense	-
2	c.3135insA / p.Asn1046fs	Frameshift	AD
3	-	-	-
4	p.Arg164fs	Frameshift	HIGH
5	c.1468C>T / p.Arg490*	Stop Gained	HIGH

Trio 2 (JAG1): Analysis detected a frameshift mutation in *JAG1* (p.Asn1046fs), consistent with the clinical presentation of **Alagille Syndrome** (Autosomal Dominant transmission). The variant was inherited from the father.

Trio 5 (MKKS): This case represents an **Autosomal Recessive (AR)** model where the child is homozygous for a nonsense mutation (p.Arg490*). The "Stop Gained" consequence results in a truncated protein, leading to **McKusick-Kaufman Syndrome**.



Visualizzazione su UCSC Genome Browser (hg38) della variante candidata nel Trio 5. La traccia VCF mostra la mutazione nonsense localizzata in un esone altamente conservato (PhyloP) del gene MKKS. Le tracce di copertura confermano la presenza del dato nel figlio e nei genitori, supportando il modello di ereditarietà autosomica recessiva per la Sindrome di McKusick-Kaufman

Trios 1 and 3 (Negative Cases): Trio 1 presented variants in *SIRPA*, but these were classified as non-pathogenic due to their moderate impact and high population frequency (MAX_AF). Trio 3 yielded no significant candidates under the *de novo* model, confirming its role as a negative control in the simulation.

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